

A circuit-differentiation framework for Green's functions on quantum computers

Samuele Piccinelli^{1,2}, Francesco Tacchino¹, Ivano Tavernelli¹, and Giuseppe Carleo²

¹IBM Research Europe – Zurich, Säumerstrasse 4, 8803 Rüschlikon, Switzerland

²Institute of Physics, École Polytechnique Fédérale de Lausanne (EPFL), CH-1015 Lausanne, Switzerland

We propose a general framework for computing Retarded Green's Functions (RGFs) on quantum computers by recasting their evaluation as a problem of circuit differentiation. Our proposal is based on real-time evolution and specifically designed circuit components, which we refer to as circuit perturbations, acting as a direct representation of the external perturbative force within the quantum circuit in a linear-response setting. The direct mapping between circuit derivatives and the computation of RGFs enables the use of a broad range of differentiation strategies. We provide two such examples, including a class of stochastic estimators which do not require extra qubit connectivity with respect to the underlying time-evolution operations. We demonstrate our approach on interacting spin and fermionic models, showing that accurate dynamical correlations can be obtained even under realistic noise assumptions. Finally, we outline how our proposal can be tied to efficient gradient-estimation techniques relevant for the fault-tolerant regime.

1 Introduction

Understanding the dynamical properties of quantum many-body systems is a central question in virtually all areas of modern physical sciences, including quantum chemistry, condensed matter and high-energy physics. In particular, the study of low-energy excitations often provides a direct bridge between theoretical models and experimental practice. However, because of the strongly correlated nature of many relevant models and the associated rapid growth of entanglement during time evolution, their investigation remains challenging at large scale. In the last decade, quantum computers have emerged as a promising computational paradigm to address this class of problems [1, 2, 3], and recent demonstrations of quantum simulations at the so-called quantum utility scale [4] have opened the way to the use of digital quantum

Samuele Piccinelli: samuele.piccinelli@ibm.com

Francesco Tacchino: fta@zurich.ibm.com

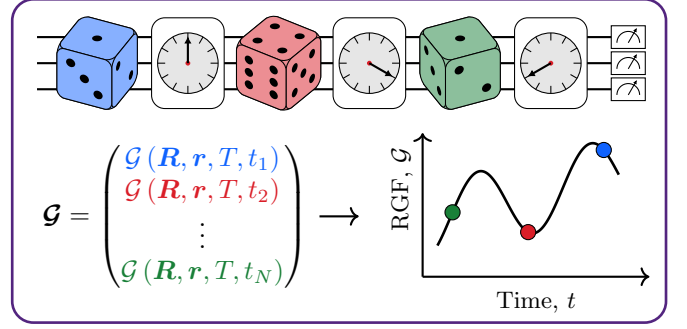


Figure 1: **Circuit perturbation method.** Our approach builds on results from linear response theory and exploits a correspondence between quantum circuits derivatives and generalized susceptibilities. By inserting randomized perturbations throughout the system evolution and leveraging stochastic gradient-estimation tools, we obtain information about RGFs at different time intervals in parallel from a single quantum circuit template.

information processing platforms as concrete research tools for the natural sciences.

In this context, time correlation functions of the form

$$\mathcal{C}(t - t_0) = \text{Tr}[\rho A(t) B(t_0)] \quad (1)$$

with $t > 0$ are particularly useful to characterize the behavior of the systems under study. In the equation above, ρ is a reference many-body state and A, B are operators associated with a particular physical process. These time correlation functions are intimately connected to the Retarded Green's Functions (RGFs), which connect theoretical predictions with experimental observables, ranging from dynamical modes and spectral functions to scattering amplitudes. The ability to compare theoretical predictions based on these functions against techniques like angle-resolved photoemission or inelastic neutron scattering constitutes a powerful tool in the hands of computational quantum scientists. Nevertheless, computing dynamical correlation functions in large-scale quantum systems remains a significant challenge for classical simulation techniques. For these reasons, quantum algorithms to compute RGFs in both the time and frequency domains have been extensively explored in the literature [5, 6, 7, 8]. Examples include quantum subspace expansion algorithms [9], a generalization of the